

PROJECT REPORT

AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

Team ID	SWTID-2026-8726
Project Name	Auto Ticket Classification using FLOW DESIGNER
Project Phase	Final Report
Date	03 October 2026

1. INTRODUCTION

1.1 Project Overview

Auto Ticket Classification using FLOW DESIGNER is a ServiceNow-based workflow automation solution designed to reduce manual effort in service-desk ticket triage. When a new Incident or Service Request is created, the workflow evaluates relevant ticket information, applies configured classification rules, assigns an appropriate category and priority, and routes the ticket to the relevant support group.

1.2 Purpose

The purpose of the project is to automate the initial classification, prioritization and routing of support tickets so that service-desk teams can reduce repetitive manual triage, improve routing consistency and process incoming tickets more efficiently.

2. IDEATION PHASE

2.1 Problem Statement

Manual ticket classification can consume service-desk time and may lead to inconsistent categorization, incorrect assignment, priority errors and unnecessary reassignment. A growing number of incoming tickets increases the workload on service-desk personnel and can delay resolution.

2.2 Empathy Map Canvas

Target users include requesters, service-desk agents, support teams, supervisors and administrators.

- Think & Feel: Concern about delay, incorrect routing and unresolved issues.
- Hear: Updates from service-desk agents and support teams about ticket status.
- See: Ticket forms, categories, priorities, assignment groups and ticket status.
- Say & Do: Submit tickets, describe issues, provide additional information and follow up.
- Pain: Manual triage, delays, reassignment and inconsistent classification.
- Gain: Faster routing, clearer ownership, reduced manual effort and more consistent handling.

2.3 Brainstorming

- Automate ticket classification using predefined keywords and conditions.
- Automatically assign ticket priority.
- Automatically route tickets to appropriate support groups.
- Provide a fallback queue for unmatched or ambiguous tickets.
- Maintain workflow logs and audit records.
- Consider future enhancements such as machine-learning classification and multilingual support.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Journey Step	Experience / System Action
Ticket Creation	Requester creates an Incident / Service Request.
Ticket Submission	Ticket information is stored in the ServiceNow record.
Automatic Trigger	Flow Designer starts when the configured record-

	created condition is satisfied.
Content Evaluation	Description, keywords, requester information and channel source are evaluated.
Classification	Decision rules determine the ticket category.
Prioritization	Configured conditions determine the ticket priority.
Routing	The ticket is assigned to the appropriate support group.
Resolution	Support team processes the ticket and resolves the issue.

3.2 Solution Requirement

FR No.	Functional Requirement	Sub Requirement
FR-1	Ticket Creation & Trigger	Create ticket and automatically trigger Flow Designer.
FR-2	Content Evaluation	Evaluate description, keywords, requester profile and channel source.
FR-3	Classification	Assign appropriate ticket category using configured conditions.
FR-4	Priority Assignment	Assign appropriate P1–P4 priority.
FR-5	Automatic Routing	Assign ticket to the appropriate support group.
FR-6	Fallback Handling	Route unmatched tickets to General Support Triage Queue.
FR-7	Audit & Monitoring	Record workflow actions, classification decisions and errors.

3.3 Data Flow Diagram

Customer / Requester → Incident / Service Request → Flow Designer Trigger → Content Evaluation & Parsing → Decision Matrix & Conditional Branching → Category + Priority + Assignment Group → Support Team → Ticket Resolution → System Logs / Audit Records.

Exception flow: Unmatched / Ambiguous Ticket → General Support Triage Queue → Manual Review.

3.4 Technology Stack

Component	Technology / Role
User Interface	ServiceNow Service Desk / ticket interface
Workflow Automation	ServiceNow Flow Designer
Ticket Data	Incident / Task records
Classification	Configured keywords and conditional rules
Routing	Assignment Group logic
User Information	ServiceNow requester/user records
Logging	ServiceNow system/audit records
Infrastructure	ServiceNow cloud platform

4. PROJECT DESIGN

4.1 Problem Solution Fit

The project addresses the gap between manual ticket triage and the need for fast, consistent ticket classification. The solution fits the problem by using Flow Designer to automate repetitive classification, priority and routing decisions while retaining a fallback path for tickets that do not match configured rules.

4.2 Proposed Solution

- Trigger the workflow automatically when a new ticket is created.
- Evaluate ticket content and relevant requester information.
- Apply decision rules to identify the ticket category.
- Assign priority according to configured conditions.
- Route the ticket to the appropriate support group.
- Send unmatched tickets to General Support Triage Queue.
- Maintain logs and audit information for traceability.

4.3 Solution Architecture

The architecture consists of Trigger & Event Ingestion, Content Evaluation & Parsing, Decision Matrix & Conditional Branching, and Assignment / Processing. The support groups can include NOC, Helpdesk, Systems Engineering and HR Operations depending on configured rules.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Focus	Story Points
Sprint-1	Ticket Intake & Content Evaluation	12
Sprint-2	Classification, Priority & Routing	16
Sprint-3	Exception Handling, Monitoring & Audit	11
Total	Complete Project	39

Planned average velocity = $39 / 3 = 13$ story points per sprint.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test Scenario	Expected Result	Result
Ticket Creation & Trigger	New ticket triggers Flow Designer	Pass
Content Evaluation	Ticket information is evaluated	Pass
Classification	Supported ticket categories are assigned	Pass
Priority Assignment	Configured priority is assigned	Pass
Assignment Routing	Correct support group is selected	Pass
Fallback Handling	Unmatched ticket reaches triage queue	Pass
Audit Logging	Workflow actions are recorded	Pass
End-to-End Flow	Ticket completes intended automated flow	Pass

Documented project test results report routing accuracy of 96.8%, average triage time of 1.2 seconds compared with 22.5 minutes previously, and SLA compliance of 98.5%.

7. RESULTS

7.1 Output Screenshots

Insert the following project screenshots in the final submission: Flow Designer workflow, ticket creation, classified ticket, priority and assignment group result, fallback queue, and audit/log output.

- Figure 1 – Flow Designer workflow
- Figure 2 – New Incident / Service Request
- Figure 3 – Automatically classified ticket
- Figure 4 – Priority and Assignment Group result
- Figure 5 – General Support Triage Queue / fallback
- Figure 6 – Workflow execution / audit output

8. ADVANTAGES & DISADVANTAGES

Advantages

- Reduces repetitive manual ticket triage.
- Provides consistent classification using configured rules.
- Automatically assigns priority and support group.
- Reduces unnecessary reassignment and routing delay.
- Provides fallback handling for unmatched tickets.
- Supports auditability through workflow and system logs.

Disadvantages

- Rule-based classification depends on the quality of configured keywords and conditions.
- Ambiguous or previously unseen ticket descriptions may require manual review.
- Rule maintenance is required when categories, terminology or routing requirements change.
- The current implementation does not include a machine-learning model.

9. CONCLUSION

Auto Ticket Classification using FLOW DESIGNER provides an automated approach to the initial handling of service tickets. By combining ticket-content evaluation, conditional classification, priority assignment and automatic routing, the solution addresses repetitive manual triage and improves the consistency of support-ticket processing. The fallback queue and audit mechanisms provide additional control for exceptions and traceability.

10. FUTURE SCOPE

- Machine-learning-based semantic ticket classification.
- Sentiment analysis for prioritization and escalation.
- Multi-language ticket classification.
- Dynamic load balancing based on agent availability and workload.
- Expansion of classification categories and business rules.

11. APPENDIX

Source Code (if any)

ServiceNow Flow Designer configuration and workflow implementation. Add exported update sets or workflow configuration files if available.

Dataset Link

Not applicable to the current rule-based implementation. Ticket records are processed within the ServiceNow environment.

GitHub & Project Demo Link

To be added by the project team if a repository or demonstration URL is available.